

Week 8 — Separating Hyperplanes and Fixed Point Theorems

Separating Hyperplane Theorem

Recall that, for $p \in \mathbb{R}^J \setminus \{0\}$ and $\alpha \in \mathbb{R}$, the hyperplane

$$H[p, \alpha] = \{x \in \mathbb{R}^J : p^\top x = \alpha\}$$

separates two sets $A, B \subseteq \mathbb{R}^J$ if, possibly after exchanging A and B ,

$$p^\top a \geq \alpha \quad \text{for every } a \in A,$$

and

$$p^\top b \leq \alpha \quad \text{for every } b \in B.$$

Theorem 1 (Separating Hyperplane Theorem). *Let $A, B \subseteq \mathbb{R}^J$ be nonempty, disjoint, convex sets. Then there exist*

$$p \in \mathbb{R}^J \setminus \{0\}, \quad \alpha \in \mathbb{R},$$

such that the hyperplane

$$H[p, \alpha] = \{x \in \mathbb{R}^J : p^\top x = \alpha\}$$

separates A and B .

Intuition. Convexity prevents either set from “wrapping around” the other. This makes it possible to find a linear boundary that places the two sets on opposite sides.

The separating hyperplane does not necessarily leave a positive distance between the sets: they may approach, or even touch, the hyperplane.

Remark (Convex hulls). Suppose $A, B \subseteq \mathbb{R}^J$ are arbitrary sets such that

$$\text{conv}(A) \cap \text{conv}(B) = \emptyset.$$

Since both convex hulls are convex, the Separating Hyperplane Theorem gives a hyperplane separating

$$\text{conv}(A) \quad \text{and} \quad \text{conv}(B).$$

Since

$$A \subseteq \text{conv}(A), \quad B \subseteq \text{conv}(B),$$

the same hyperplane also separates A and B .

Fixed Point Theorems

Fixed Points of Functions

Definition (Fixed point). Let

$$f : X \rightarrow X.$$

A point $x^* \in X$ is a *fixed point* of f if

$$\boxed{f(x^*) = x^*}.$$

We say that f has the *fixed point property* if there exists at least one $x^* \in X$ satisfying this equality.

Intuition. A fixed point is a point that does not move under the map. We input x^* into f and obtain exactly the same point back.

Basic Examples

Example. Consider

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = x.$$

Then

$$f(x) = x \quad \forall x \in \mathbb{R},$$

so every point is a fixed point:

$$\text{Fix}(f) = \mathbb{R}.$$

Example. Let

$$g : \mathbb{R} \rightarrow \mathbb{R}, \quad g(x) = x + 1.$$

A fixed point would satisfy

$$x + 1 = x,$$

which is impossible. Hence

$$\text{Fix}(g) = \emptyset.$$

Example. Let

$$h : \mathbb{R} \rightarrow \mathbb{R}, \quad h(x) = -x.$$

A fixed point satisfies

$$-x = x,$$

so

$$x = 0.$$

Hence

$$\text{Fix}(h) = \{0\}.$$

Remark. Whether a map has a fixed point depends crucially not only on its formula but also on its domain and codomain.

Why the Domain Matters

Consider

$$f(x) = 2x.$$

As a map

$$f : \mathbb{R} \rightarrow \mathbb{R},$$

the point

$$x^* = 0$$

is a fixed point.

However, consider the restriction

$$f : [1, \infty) \rightarrow [1, \infty).$$

The fixed-point equation remains

$$2x = x,$$

whose only solution is

$$x = 0.$$

But

$$0 \notin [1, \infty),$$

so this restricted map has no fixed point.

Intuition. A solution to the algebraic equation

$$f(x) = x$$

is relevant only if it belongs to the set on which the map is defined. Restricting the domain can therefore destroy fixed points.

Examples Motivating Brouwer's Theorem

The following examples show that continuity, compactness, and convexity will all play important roles.

Failure of Continuity

Define

$$f : [0, 1] \rightarrow [0, 1]$$

by

$$f(x) = \begin{cases} 1, & x < 1, \\ 0, & x = 1. \end{cases}$$

The set $[0, 1]$ is compact and convex, but f is discontinuous. There is no fixed point. Indeed:

- if $x < 1$, then

$$f(x) = 1 \neq x;$$

- if $x = 1$, then

$$f(1) = 0 \neq 1.$$

Thus compactness and convexity alone do not guarantee a fixed point.

Failure of Closedness / Compactness

Define

$$f : [0, 1) \rightarrow [0, 1)$$

by

$$f(x) = \frac{x+1}{2}.$$

The map is continuous and the domain is bounded and convex, but

$$[0, 1)$$

is not closed and hence is not compact.

A fixed point would satisfy

$$\frac{x+1}{2} = x,$$

so

$$x = 1.$$

But

$$1 \notin [0, 1).$$

Hence the map has no fixed point.

Failure of Boundedness

Consider

$$f : [1, \infty) \rightarrow [1, \infty), \quad f(x) = 2x.$$

The domain is closed and convex, and f is continuous, but the domain is not bounded and hence not compact.

The only solution of

$$2x = x$$

is

$$x = 0,$$

which is not in the domain. Therefore there is no fixed point.

Failure of Convexity

Let

$$X = \{-1, 1\}$$

and define

$$f : X \rightarrow X$$

by

$$f(x) = -x.$$

The set X is compact, and f is continuous, but X is not convex.

Moreover,

$$f(-1) = 1, \quad f(1) = -1,$$

so f has no fixed point.

Remark. These examples separately illustrate the roles of the assumptions appearing in Brouwer's theorem:

continuity + compactness + convexity.

Brouwer's Fixed Point Theorem in One Dimension

Theorem 2 (Brouwer Fixed Point Theorem in $\mathbb{R}$). *Let*

$$f : [a, b] \rightarrow [a, b]$$

be continuous. Then f has a fixed point: there exists

$$x^* \in [a, b]$$

such that

$$f(x^*) = x^*.$$

Proof. If

$$f(a) = a$$

or

$$f(b) = b,$$

then we already have a fixed point.

Suppose instead that neither endpoint is fixed. Since

$$f(a) \in [a, b],$$

we must have

$$f(a) > a.$$

Likewise, since

$$f(b) \in [a, b],$$

we must have

$$f(b) < b.$$

Define

$$g : [a, b] \rightarrow \mathbb{R}$$

by

$$g(x) = f(x) - x.$$

Because f is continuous, g is continuous.

Moreover,

$$g(a) = f(a) - a > 0,$$

while

$$g(b) = f(b) - b < 0.$$

By the Intermediate Value Theorem, there exists

$$x^* \in (a, b)$$

such that

$$g(x^*) = 0.$$

Therefore

$$f(x^*) - x^* = 0,$$

and hence

$$\boxed{f(x^*) = x^*}.$$

□

Intuition. Plot $f(x)$ together with the 45° line

$$y = x.$$

A fixed point is an intersection of the graph of f with this line.

Continuity and the condition

$$f([a, b]) \subseteq [a, b]$$

force at least one such intersection.

Brouwer's Fixed Point Theorem

Theorem 3 (Brouwer Fixed Point Theorem). *Let*

$$X \subseteq \mathbb{R}^J$$

be nonempty, compact, and convex. If

$$f : X \rightarrow X$$

is continuous, then there exists

$$x^* \in X$$

such that

$$\boxed{f(x^*) = x^*}.$$

Remark. The one-dimensional theorem above is the special case

$$X = [a, b] \subseteq \mathbb{R}.$$

Closed Graphs and Continuity

Recall that the graph of a function

$$f : X \rightarrow Y$$

is

$$\text{Gr}(f) = \{(x, y) \in X \times Y : y = f(x)\}.$$

Theorem 4 (Closed Graph Theorem for Functions). *Let (X, d_X) and (Y, d_Y) be metric spaces and let*

$$f : X \rightarrow Y.$$

1. *If f is continuous, then*

$$\text{Gr}(f)$$

is closed in $X \times Y$.

2. *If Y is compact and*

$$\text{Gr}(f)$$

is closed in $X \times Y$, then f is continuous.

Remark. The compactness assumption on the codomain matters for the converse. A closed graph alone does not guarantee continuity in complete generality.

Brouwer via Closed Graphs

Combining the Closed Graph Theorem with Brouwer gives the following useful formulation.

Corollary. Let

$$X \subseteq \mathbb{R}^J$$

be nonempty, compact, and convex. If

$$f : X \rightarrow X$$

has a closed graph, then f has a fixed point.

Proof. Since X is compact and

$$\text{Gr}(f)$$

is closed, the Closed Graph Theorem implies that f is continuous.

Brouwer's Fixed Point Theorem then guarantees a fixed point. $\square$

Fixed Points of Correspondences

Correspondences

A correspondence allows each input to be associated with a set of possible outputs rather than with a single point.

We write

$$F : X \rightrightarrows Y$$

to indicate that

$$F(x) \subseteq Y$$

for every $x \in X$.

A function

$$f : X \rightarrow Y$$

can be viewed as the special correspondence

$$F(x) = \{f(x)\}.$$

Definition (Fixed point of a correspondence). Let

$$F : X \rightrightarrows X.$$

A point $x^* \in X$ is a *fixed point* of F if

$$x^* \in F(x^*).$$

Remark. For a function, the fixed-point condition is

$$f(x^*) = x^*.$$

For a correspondence, we do *not* require

$$F(x^*) = \{x^*\}.$$

We only require the point itself to belong to its image:

$$x^* \in F(x^*).$$

Graph of a Correspondence

Definition (Graph). The graph of

$$F : X \rightrightarrows Y$$

is

$$\text{Gr}(F) = \{(x, y) \in X \times Y : y \in F(x)\}.$$

Why Convex-Valuedness Matters

Consider the correspondence

$$F : [0, 1] \rightrightarrows [0, 1]$$

defined by

$$F(x) = \begin{cases} \{1\}, & x < \frac{1}{2}, \\ \{0, 1\}, & x = \frac{1}{2}, \\ \{0\}, & x > \frac{1}{2}. \end{cases}$$

There is no fixed point:

- if $x < 1/2$, then

$$F(x) = \{1\},$$

so $x \notin F(x)$;

- if $x > 1/2$, then

$$F(x) = \{0\},$$

so again $x \notin F(x)$;

- at $x = 1/2$,

$$F(1/2) = \{0, 1\},$$

and therefore

$$\frac{1}{2} \notin F(1/2).$$

Its graph is

$$\text{Gr}(F) = \left[0, \frac{1}{2}\right] \times \{1\} \cup \left[\frac{1}{2}, 1\right] \times \{0\},$$

which is closed.

Hence closedness of the graph is not the problem.

The problem is that

$$F\left(\frac{1}{2}\right) = \{0, 1\}$$

is not convex.

Kakutani's Fixed Point Theorem: Closed-Graph Version

Definition (Convex-valued correspondence). A correspondence

$$F : X \rightrightarrows Y$$

is *convex-valued* if

$$F(x)$$

is a convex subset of Y for every $x \in X$.

Definition (Nonempty-valued correspondence). A correspondence

$$F : X \rightrightarrows Y$$

is *nonempty-valued* if

$$F(x) \neq \emptyset \quad \text{for every } x \in X.$$

Theorem 5 (Kakutani Fixed Point Theorem: closed-graph version). *Let*

$$X \subseteq \mathbb{R}^J$$

be nonempty, compact, and convex.

Suppose

$$F : X \rightrightarrows X$$

is

1. *nonempty-valued;*
2. *convex-valued;*
3. *and has a closed graph.*

Then there exists

$$x^* \in X$$

such that

$$\boxed{x^* \in F(x^*)}.$$

Intuition. Kakutani generalizes Brouwer from functions to correspondences.

If every value is a singleton,

$$F(x) = \{f(x)\},$$

then convex-valuedness is automatic and the correspondence fixed-point condition

$$x^* \in F(x^*)$$

reduces to

$$f(x^*) = x^*.$$

Continuity of Correspondences

Upper and Lower Inverses

For a correspondence, there are two natural ways to define an inverse image of a set.

Let

$$F : X \rightrightarrows Y$$

and let

$$E \subseteq Y.$$

Definition (Upper inverse). The *upper inverse* of E under F is

$$\boxed{F^u(E) = \{x \in X : F(x) \subseteq E\}}.$$

Definition (Lower inverse). The *lower inverse* of E under F is

$$F^\ell(E) = \{x \in X : F(x) \cap E \neq \emptyset\}.$$

Intuition. The two inverses answer different questions.

The upper inverse asks:

For which x does the *entire* set $F(x)$ lie inside E ?

The lower inverse asks:

For which x does $F(x)$ contain *at least one* point in E ?

Upper and Lower Hemicontinuity

Let (X, d_X) and (Y, d_Y) be metric spaces.

Definition (Upper hemicontinuity). A correspondence

$$F : X \rightrightarrows Y$$

is *upper hemicontinuous* if, for every open set

$$E \subseteq Y,$$

the upper inverse

$$F^u(E) = \{x \in X : F(x) \subseteq E\}$$

is open in X .

Definition (Lower hemicontinuity). A correspondence

$$F : X \rightrightarrows Y$$

is *lower hemicontinuous* if, for every open set

$$E \subseteq Y,$$

the lower inverse

$$F^\ell(E) = \{x \in X : F(x) \cap E \neq \emptyset\}$$

is open in X .

Functions as Singleton-Valued Correspondences

Let

$$f : X \rightarrow Y$$

be a function and define the associated correspondence

$$F(x) = \{f(x)\}.$$

Then

$$F^u(E) = \{x : \{f(x)\} \subseteq E\} = \{x : f(x) \in E\} = f^{-1}(E).$$

Likewise,

$$F^\ell(E) = \{x : \{f(x)\} \cap E \neq \emptyset\} = \{x : f(x) \in E\} = f^{-1}(E).$$

Therefore, for singleton-valued correspondences,

$$F^u(E) = F^\ell(E) = f^{-1}(E).$$

Hence

$$f \text{ is continuous} \iff F \text{ is both upper and lower hemicontinuous.}$$

Example: Upper but Not Lower Hemicontinuity

Define

$$F : [0, 1] \rightrightarrows [0, 1]$$

by

$$F(x) = \begin{cases} \{0\}, & x < 1, \\ [0, 1], & x = 1. \end{cases}$$

All notions of openness below use the relative topology on $[0, 1]$.

We claim that F is upper hemicontinuous.

Let $E \subseteq [0, 1]$ be open.

- If

$$E = [0, 1],$$

then

$$F^u(E) = [0, 1].$$

- If

$$0 \in E \quad \text{and} \quad E \neq [0, 1],$$

then

$$F^u(E) = [0, 1).$$

- If

$$0 \notin E,$$

then

$$F^u(E) = \emptyset.$$

All three sets are open in the relative topology of $[0, 1]$. Hence F is upper hemicontinuous.

However, F is not lower hemicontinuous.

Take the relatively open set

$$E = (0, 1].$$

Then

$$F^\ell(E) = \{1\},$$

because only

$$F(1) = [0, 1]$$

intersects $(0, 1]$.

But

$$\{1\}$$

is not open in $[0, 1]$. Therefore F is not lower hemicontinuous.

Its graph is

$$\text{Gr}(F) = ([0, 1] \times \{0\}) \cup (\{1\} \times [0, 1]),$$

which is closed.

Example: Lower but Not Upper Hemicontinuity

Now define

$$F : [0, 1] \rightrightarrows [0, 1]$$

by

$$F(x) = \begin{cases} [0, 1], & x < 1, \\ \{0\}, & x = 1. \end{cases}$$

This correspondence is lower hemicontinuous.

Indeed, let $E \subseteq [0, 1]$ be open.

If

$$E = \emptyset,$$

then

$$F^\ell(E) = \emptyset.$$

If

$$0 \in E,$$

then every $F(x)$ intersects E , so

$$F^\ell(E) = [0, 1].$$

If

$$E \neq \emptyset \quad \text{and} \quad 0 \notin E,$$

then

$$F^\ell(E) = [0, 1),$$

which is open in the relative topology.

Thus F is lower hemicontinuous.

However, it is not upper hemicontinuous.

Take

$$E = [0, 1),$$

which is open in the relative topology on $[0, 1]$.

For every $x < 1$,

$$F(x) = [0, 1] \not\subseteq E,$$

while

$$F(1) = \{0\} \subseteq E.$$

Hence

$$F^u(E) = \{1\},$$

which is not open.

Therefore F is not upper hemicontinuous.
The graph is not closed. For example, if

$$x_n \uparrow 1$$

and

$$y_n = 1,$$

then

$$(x_n, 1) \in \text{Gr}(F)$$

for every n , but

$$(x_n, 1) \rightarrow (1, 1),$$

while

$$1 \notin F(1) = \{0\}.$$

Thus

$$(1, 1) \notin \text{Gr}(F).$$

Closed Graph Theorem for Correspondences

Definition (Closed-valued correspondence). A correspondence

$$F : X \rightrightarrows Y$$

is *closed-valued* if

$$F(x)$$

is closed in Y for every $x \in X$.

Theorem 6 (Closed Graph Theorem for Correspondences). *Let (X, d_X) and (Y, d_Y) be metric spaces and suppose that Y is compact.*

Then a correspondence

$$F : X \rightrightarrows Y$$

has a closed graph if and only if

1. *F is upper hemicontinuous; and*
2. *F is closed-valued.*

Intuition. Upper hemicontinuity prevents the correspondence from suddenly “expanding” as the input changes.

Closed-valuedness prevents limit points of $F(x)$ from falling outside the value set itself.

When the codomain is compact, these two properties together are exactly what is needed for the graph to be closed.

Kakutani's Fixed Point Theorem

Combining the closed-graph characterization with the previous version of Kakutani gives the standard hemicontinuity formulation.

Theorem 7 (Kakutani Fixed Point Theorem). *Let*

$$X \subseteq \mathbb{R}^J$$

be nonempty, compact, and convex.

Suppose

$$F : X \rightrightarrows X$$

satisfies:

1. *F is nonempty-valued:*

$$F(x) \neq \emptyset \quad \forall x \in X;$$

2. *F is convex-valued:*

$$F(x) \text{ is convex} \quad \forall x \in X;$$

3. *F is closed-valued:*

$$F(x) \text{ is closed} \quad \forall x \in X;$$

4. *F is upper hemicontinuous.*

Then F has a fixed point: there exists

$$x^* \in X$$

such that

$$\boxed{x^* \in F(x^*)}.$$

Remark. Because X is compact, every closed value

$$F(x) \subseteq X$$

is itself compact.

Thus Kakutani is also often stated by assuming that F is nonempty, compact-valued, convex-valued, and upper hemicontinuous.